

OPFsembleR: An Optimum-Path Forest-based Framework for Ensemble Pruning

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Abstract. Selecting machine learning algorithms that properly fit the data distribution under analysis is a key challenge in classification and regression tasks. The standard strategy for this problem is the well-known ensemble learning, which combines the results of distinct models to handle a given classification task. Such an approach is costly because it requires merging model predictions and ensuring consistent variability across results, especially as the number of baselines in the ensemble increases. Clustering algorithms efficiently tackle this problem by grouping similar models according to their outcome predictions. This paper proposes OPFsembleR, an extension of the OPFsemble approach for regression models that employs the Optimum-Path Forest framework to learn the most important models and further prune less representative ones in an unsupervised fashion while preserving ensemble diversity. The proposed study also introduces two voting strategies to minimize bias and variance when selecting models from each group. Results across several public datasets are compared using the stacked generalization strategy to assess the model’s performance, revealing that the proposed ensemble pruning strategy achieves the best scores and shows statistical similarity to the baselines that employ the full ensemble.

Keywords: Ensemble Pruning · Optimum-Path Forest · Ensemble Learning

1 Introduction

Machine learning methods are generally guided by the properties of the data distribution under analysis, which affect results based on data variability and class separation. However, many methods struggle with differences between training and test data. A popular strategy for this problem is ensemble learning, which combines multiple models to reduce bias and variance, thereby enhancing prediction accuracy. Notable strategies include *Bagging* [22], which uses distinct data subsets for training; *Boosting* [32], which assigns weights to models based on performance; and *stacked generalization* [39], where a meta-learner combines predictions from various classifiers to improve overall outcomes.

While ensemble methods improve accuracy by promoting *diversity* and reducing bias and variance, they are sometimes limited by the *complexity* of their baseline mod-

els. Therefore, minimizing the number of models while maximizing diversity and predictive performance becomes critical in ensemble learning. *Ensemble pruning* addresses this shortcoming by *reducing* complexity and computational cost during training. Pruning can be achieved by modeling the problem as an optimization task called *classifier selection* or *classifier weighting*, in which each model in the ensemble is assigned a weight. The former is a specific case in which the weights are either 1 or 0.

Clustering classifiers is an interesting approach to identifying the most relevant ones, as we assume that models within the same cluster are similar, thereby not promoting diversity. We, therefore, need a heuristic to select one from each cluster to compose the final (pruned) ensemble. Such an idea is not new, but here we focus on methodologies that employ the Optimum-Path Forest (OPF) to fulfill that purpose. OPF is a framework that comprises supervised [28,27,29], semi-supervised [3], and unsupervised [31] techniques for pattern recognition based on graph partitions. A particular feature of the unsupervised version is its ability to estimate the number of clusters on the fly, which is extremely helpful when this information is unknown beforehand.

Previous work has considered OPF-based classifiers for ensemble pruning. Fernandes and Papa [25] used hypercomplex optimization to find the smallest subset of classifiers that maximizes the ensemble’s effectiveness. Jodas et al. [19] introduced a strategy based on Graph Matching Networks and Gaussian Mixture Models to prune OPF ensembles, though it is limited to OPF classifiers. In another work, Jodas et al. [20] presented OPFsemble, which employs unsupervised OPF to group classification models with similar behavior and selects the best from each cluster, thereby promoting diversity and heterogeneous ensembles, reducing complexity, and maintaining accuracy. Here, we extend their work by proposing OPFsembleR for regression models and two more voting strategies to minimize bias and variance when selecting models within each cluster.

Primary contributions. This paper provides four significant contributions: i) an ensemble pruning framework for heterogeneous classification and regression tasks; ii) the application of two voting strategies to select the best-scored ensemble models and combine their set of predictions; iii) the use of several divergence measures to assess the robustness of the proposed approach against baselines; and iv) to promote state-of-the-art research related to OPF pruning strategies.

The remainder of this paper is organized as follows: Section 2 presents a theoretical background of the Optimum-Path Forest framework, and Section 3 presents the proposed method in detail. The datasets used and experimental setup are presented in Section 4, while Section 5 discusses the results. Finally, Section 6 states conclusions and future works.

2 Unsupervised Learning with Optimum-Path Forest

The unsupervised OPF encodes each training instance as a graph node, where weighted edges connect each sample to its k -nearest neighbors based on a distance or similarity function. The nodes are also weighted according to a probability density function (PDF) computed as follows:

$$\rho(s) = \frac{1}{\sqrt{2\pi\psi^2k}} \sum_{\forall z \in \mathcal{N}_k(s)} \exp\left(\frac{-d^2(s,z)}{2\psi^2}\right), \quad (1)$$

where $\mathcal{N}_k(s)$ stands for the k -neighborhood of sample s , $\psi = \frac{m_w}{3}$ such that m_w is the maximum weight among all edges in the graph, and d is a function that computes the distance or similarity among two nodes. The parameter ψ is selected to incorporate all adjacent nodes in the density computation, as the Gaussian function captures the majority of samples within the range $d(s,z) \in [0, 3\psi]$.

The probability density function is computed by estimating the optimum number of nearest neighbors $k^* \in [1, k_{max}]$ that minimizes the graph cut over the training set $\mathcal{T}$, where k_{max} is the maximum value for the nearest neighbors k^* . The greater the value of k_{max} , the more connected the resulting graph will generally be. There is no strict guideline for choosing k_{max} ; however, for large datasets, we suggest setting it to about 1% of the dataset size to prevent excessive computational cost. The value of k^* is usually set to $[1, k_{max}]$ by means of hyperparameter tuning. Finally, the unsupervised OPF defines a set of *prototypes* $\mathcal{P}$ that comprises one element per maximum of the PDF. In the sequence, a sample z is designated to the path whose minimum density value along it is maximum, as follows:

$$\begin{aligned} f_{min}(\langle z \rangle) &= \begin{cases} \rho(z) & \text{if } z \in \mathcal{P} \\ \rho(z) - \epsilon & \text{otherwise,} \end{cases} \\ f_{min}(\langle \phi_s \cdot \langle s, z \rangle \rangle) &= \min\{f_{min}(\phi_s), \rho(z)\}, \end{aligned} \quad (2)$$

where ϵ is a small constant used to place prototypes in the peaks of the PDF, and ϕ_s denotes the sequence of neighboring samples, that is, a path that begins at a root in $\mathcal{P}$ and ends at the sample s . Figure 1 illustrates a toy example of the OPF working mechanism on the boat dataset¹.

Essentially, OPF builds an optimum-path forest in which each optimum-path tree comprises a cluster of the most strongly connected samples. Each cluster figures a prototype, which is the root of the tree. If one picks any sample from a given tree and follows the directed arcs, it will end up in the root of that tree.

3 Proposed method

OPFsembleR² groups mixed (heterogeneous) models that perform similarly when classifying some particular dataset. The method uses unsupervised OPF to cluster and select the *most representative models* by exploring the entire cluster space to maximize *ensemble diversity*. We initially assemble a set of n -diverse models and use k -fold cross-validation on the dataset $\mathcal{V}$ to obtain predictions from each classifier in the ensemble.

¹Available at: <https://github.com/gugarosa/opfpython/tree/main/data>

²Code available at: <https://github.com/danilojodas/OpfSembleRegression/tree/master>

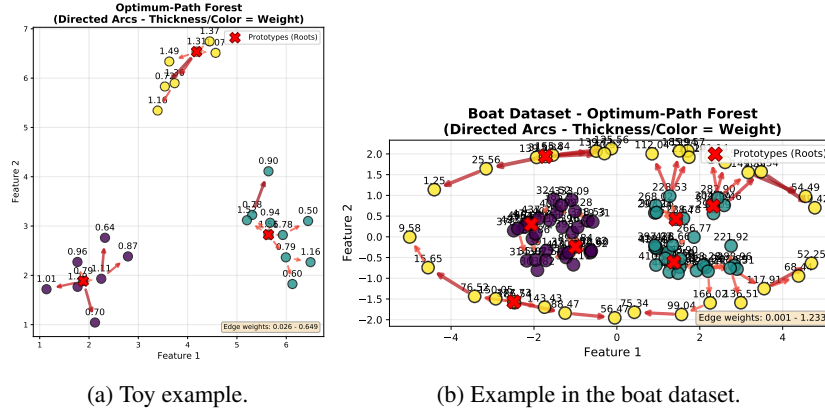


Fig. 1: Illustration of the directed arcs connecting the samples within the clusters. Note the density over the nodes computed using Equation 1, where the highest values are assigned to the prototypes (red crosses).

The set $\mathcal{V}$ is split into two subsets $\mathcal{V}_1$ and $\mathcal{V}_2$ for training and validation of the base models at each cross-validation fold, respectively.

The predictions obtained from each validation set $\mathcal{V}_2$ are stacked to create the meta-data set $\mathcal{S}$ of n samples and c features, where c is the number of predictions obtained from the validation sets $\mathcal{V}_2$ across all cross-validation folds.

The samples in the set $\mathcal{V}_2$ differ across cross-validation folds, resembling a bootstrap aggregation without replacement across all cross-validation iterations. Finally, the set $\mathcal{S}$ is used to assemble clusters of similar models via the unsupervised OPF, where the prototype of each cluster is the model that will compose the final ensemble. Figure 2 illustrates the OPFsembleR working mechanism for ten mixed models and the clustering according to their similar predictions in the validation sets.

Classification OPFsembleR provides a pruning mechanism designed for supervised classification tasks. In this approach, each metadata element P_{ij} reflects the successes and failures of the i -th model for each sample j from the k -fold cross-validation set (check Figure 2 for more details). A value of 0 denotes a misclassified sample, while 1 indicates correct classification. Each classifier is assigned the F1-score computed from its predictions on all validation set samples in each cross-validation fold. Such scores are used in subsequent assessments to select the most representative models during voting-based prediction. The predictive step assumes the best-performing combination of classifiers identified via a specific selection criterion. Such a strategy ensures ensemble diversity tailored by different representative classifiers. Figure 3 depicts the proposed voting strategies to combine the final predictions: (a) **Proto**: the most common predicted class among the prototypes; and (b) **Intracluster**: the most common predicted class among predictions from the highest-scored cluster, i.e., the cluster with the best-performing baseline classifiers.

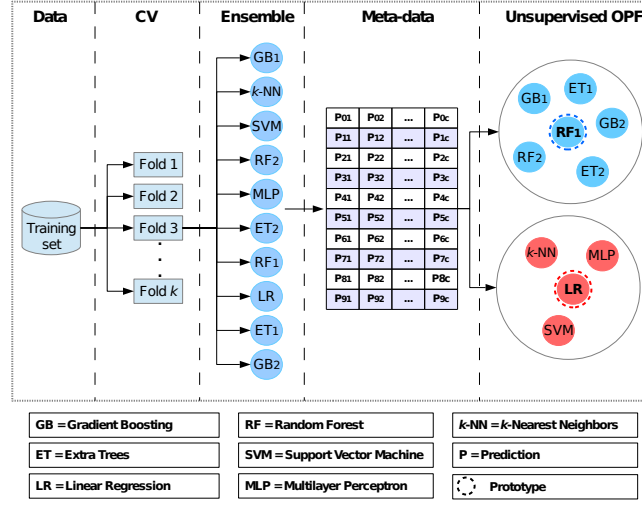


Fig. 2: OPFsembleR working mechanism.

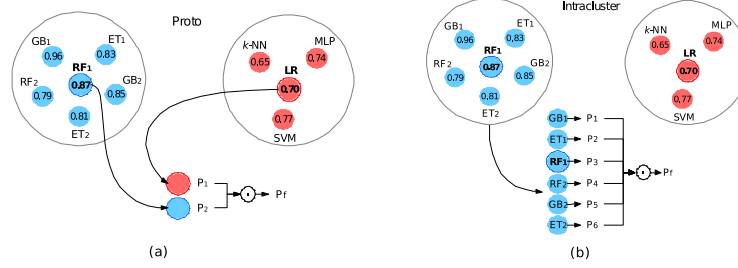


Fig. 3: Illustration of the voting approaches of the OPFsemble: (a) Proto voting and (b) Intracluster voting. Notice that P_f is the final prediction and $\odot$ denotes the combination of predictions provided by each selected classifier, respectively. The prototype node is the one surrounded by a solid line.

Ensemble diversity enhances performance by correcting errors among individual ensemble members, thereby reducing total error and improving accuracy. Wood et al. [40] show that balancing diversity with accuracy improves generalization and robustness, while Wu et al. [41] find that higher diversity metrics correlate with increased accuracy in vision tasks. In this sense, OPFsembleR promotes diversity through the *Proto* strategy, which selects one prototype model from each cluster identified by the unsupervised OPF method. This selection ensures diverse decision patterns and adheres to the principle of combining heterogeneous models for better generalization. The clustering step groups models based on similarities in their predictions to avoid redundancy while maintaining heterogeneity.

To address the increased computational cost associated with the dimensionality of predictions in the unsupervised OPF algorithm, we evaluated three diversity measures

to optimize metadata representation: Yule’s Q, Disagreement, and Class Count. Yule’s Q estimates the association between classifiers, ranging from -1 (perfect negative association) to 1 (perfect positive association). This measure is suitable for binary variables. The Disagreement measure assesses how frequently pairs of classifiers disagree, based on the variance or entropy of their predictions. In the context of binary tasks, it reflects the pairwise error rates. Finally, Class Count, as proposed by Candido et al. [7], creates a matrix that counts how often each classifier predicts each class. This approach further reduces the metadata’s dimensionality. Each diversity measure is detailed in the subsections below.

Regression The regression task performs similarly to the strategy adopted for supervised classification in the clustering and pruning procedures used to select the most representative regression models. At first glance, the ensemble is initialized with n mixed models configured with different hyperparameter setups. Afterward, each model is trained on the training set $\mathcal{V}_1$ using a cross-validation procedure to create subsequent metadata, using stacked predictions from each validation set $\mathcal{V}_2$. In addition, each model receives the average Mean Absolute Error (MAE) after performing the predictive step over the validation samples.

Subsequently, we employed a similar approach to reduce the metadata size across the feature dimension describing the set of validation sample predictions. Unlike classification tasks, where diversity among models is often measured by discrete hits and misses, regression evaluation relies on continuous measures that capture the proximity of predictions to the true values. In this context, this process involves applying the Pearson correlation coefficient to the set of metadata predictions to capture the pairwise similarity among the employed models. This measure is particularly well-suited to the proposed problem, as it directly assesses the similarity of forecasts at different points in the validation set.

Further, we reduce the dimensionality of the original prediction matrix $n \times c$ into a compact matrix $n \times n$. Such a strategy allows the unsupervised OPF to identify those exhibiting similar performance under a novel metadata representation. Based on correlations between pairs of models, this transformation preserves critical information about performance similarity, enabling the OPF algorithm to identify natural groupings and select the most representative models in each cluster. This process significantly reduces the computational cost of handling large volumes of metadata.

4 Dataset and Experimental Setup

Datasets In both tasks, i.e., classification and regression, the data underwent preprocessing to address missing values, in which missing values in each feature were imputed with the corresponding average value. In addition, each categorical attribute underwent a sequential transformation into an integer representation. Furthermore, to standardize the data, all values were scaled based on the assumption of a Gaussian distribution. The following datasets were applied in this work:

Classification: Experiments were performed over 15 datasets from the UCI Machine Learning Repository [12]: Cervical Cancer (CC) [15], Diabetic Retinopathy Debreccen (DRD) [4], Forest Type Mapping (FTM) [21], Glass Identification (GI) [17], Indian Liver (IL) [30], Iris (IRS) [16], Mammographic Mass (MM) [13], Primary Tumor (PT) [13], Seismic Bumps (SB) [33], Speaker Accent Recognition (ACCT) [1], Statlog Vehicle Silhouette (SVS) [24], Vertebral Column (VC) [5], Wisconsin Breast Cancer Original (BCO) [23], Wisconsin Breast Cancer Diagnostic (BCD) [37] and Wisconsin Breast Cancer Prognostic (BCP) [38].

Regression: The regression pruning performance was assessed using 15 publicly available Kaggle and UCI Machine Learning Repository datasets: Airfoil Self-Noise (ASN) [6], Concrete Compressive Strength (CCS) [42], Distillation Column (DC)^{3,4}, Energy Efficiency (EE)⁵[35], Effluent Treatment Plant (ETP)⁶, Forest Fires (FF) [11], Power Transformers Health Condition (PTHC)⁷ [26], Mosquito Indicator (MI)⁸, Parkinson’s Telemonitoring (PT) [34], QSAR Aquatic Toxicity (QAT) [8], QSAR Fish Toxicity (QFT) [9], Student Performance (SP)⁹ [10], Synchronous Machine (SM) [2], Tree Risk Probability (TRP) [7], and Yacht Hydrodynamics (YH) [18].

Experimental setup OPFsembleR was compared with the stacked generalization ensemble [36], k -Means, and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [14] to evaluate its performance against a strategy that uses the full set of baseline classifiers and similar clustering approaches. Each dataset was split into 20 folds, with 70% for training, 15% for validation, and 15% for testing.

To build the ensemble set $\mathcal{S}$, we set $k = 10$ for predictions from each training subset within the 20-fold splits. For the unsupervised OPF step, we optimized k_{max} via Grid Search over [5, 10, 20, 30, 40, 50] to select the value that maximizes the F1-score on the validation set. This range was also used for k -Means and DBSCAN. It’s important to note that k_{max} controls local connectivity in the OPF, rather than representing the number of clusters, allowing final clusters to emerge from data distribution.

OPFsembleR comprises 10, 30, and 50 instances of Extra Trees, Gradient Boosting, k -Nearest Neighbors, MLP, Naive Bayes, Random Forest, and SVM, each with randomized hyperparameters, for both classification and regression tasks. The same setup was used for stacked generalization, k -Means, and DBSCAN, with the supervised OPF classifier included in the classification ensembles but not in the regression ensembles, as OPF does not support regression tasks.

For DBSCAN and k -Means, we adopt the proto and intracluster strategies, defining centroids as the observations closest to each cluster’s mean, computed using Euclidean

³<https://www.kaggle.com/datasets/amarhaiqal/aspen-hysys-distillation-column-data>

⁴Prediction of the Heat Exchanger (HX) and Temperature Indicator (TX).

⁵Prediction of the heating and cooling loading.

⁶<https://www.kaggle.com/datasets/kane6543/etp-dataset>

⁷<https://www.kaggle.com/datasets/easonlai/sample-power-transformers-health-condition-dataset>

⁸<https://www.kaggle.com/datasets/kukuroo3/mosquito-indicator-in-seoul-korea>

⁹Prediction of the Portuguese (P), Mathematic (M) and final year grade (G3) scores.

distances. This approach generates estimated centroids for the "proto" strategy. Performance was assessed using F1 Score for classification and Mean Absolute Error for regression.

5 Results

5.1 Classification

This section presents a comparison between OPFsembleR and stacked generalization. Figure 4 shows the average scores obtained by both ensemble methods, i.e., OPF intracluster and OPF proto, and each diversity metric applied to OPFsembleR. In most cases, OPFsembleR achieved scores comparable to, and in some cases higher than, those of the stacking ensemble approach. This behavior is confirmed in the Diabetic Retinopathy Debrecen (DRD), Glass Identification (GI), and Statlog Vehicle Silhouette (SVS) datasets for OPF intracluster, as well as in the Speaker Accent Recognition (ACCT), Breast Cancer Prognostic (BCP), and Vertebral Column (VC) datasets for OPF Proto. The results may vary with the number of baseline classifiers used in each ensemble technique.

The results in Figure 4 illustrate the impact of ensemble diversity on predictive performance. The *Proto* strategy achieves greater diversity by selecting models from distinct clusters, whereas the *Intracluster* approach focuses on classifiers within the same high-performing group. This distinction is particularly evident in datasets such as *Indian Liver (IL)* and *Cervical Cancer (CC)*, where the prototype-based ensemble achieved higher F1 scores. These results suggest that including classifiers with complementary behaviors can enhance the ensemble’s effectiveness. Moreover, the small performance differences observed across the diversity measures, namely Yule’s Q, Disagreement, and *Class Count*, demonstrate the robustness of the OPF-based clustering process and indicate that ensemble diversity is preserved regardless of the specific metric used for metadata reduction.

Figure 5 shows that OPFSembleR outperformed all the other techniques in some cases, i.e., the IL dataset, while obtaining similar results in most of the other cases. Moreover, OPFSembleR outperformed the stacking approach in terms of F1-score in most cases, particularly for the PT, ACCT, and GI datasets. It is worth noting that OPFSembleR’s main advantage lies in its ability to reduce ensemble size while maintaining competitive predictive performance. Stacking combines all base classifiers, which can lead to redundancy and increased computational cost, especially during inference. In contrast, OPFsembleR prunes the ensemble by selecting representative classifiers via clustering, yielding a more compact solution. This trade-off is particularly valuable in applications where inference time, memory usage, or model interpretability are critical constraints.

5.2 Regression

In this subsection, we analyze the performance of the proposed regression model, considering different pruning approaches and the impact of divergence on model selection.

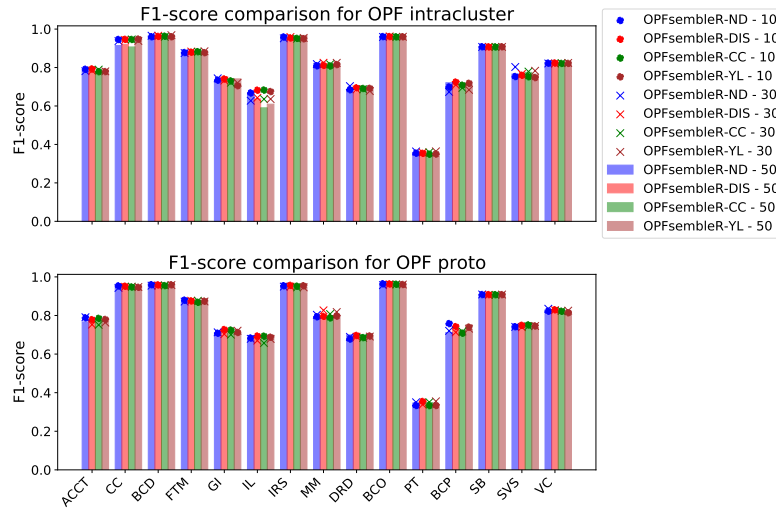


Fig. 4: Average F1-score obtained from each ensemble model and diversity measure. ND, CC, DIS, and YL stand for Non-Divergence, *Class Count*, disagreement, and Yule's, respectively. The numbers following each label indicate the number of classifiers in the ensemble setup.

The results for 10, 30 and 50 classifiers are presented in Tables 1, 2 and 3, respectively, which summarizes the average MAE and standard deviation obtained for each dataset and experimental configuration. Note that the bold values denote the best-performing models' results.

Table 1: Average MAE for each dataset and ensemble approach for 10 classifiers.

	DBSCAN proto	DBSCAN intracluster	k-Means proto	k-Means intracluster	OPF proto	OPF intracluster	Stacking
ASN	3.449±0.329	2.403±0.163	2.298±0.216	1.164±0.139	1.414±0.149	1.230±0.156	1.021±0.125
DC - HX	0.000±0.000	0.000±0.000	0.002±0.001	0.000±0.000	0.000±0.000	0.000±0.000	0.002±0.000
DC - TX	0.000±0.000	0.000±0.000	0.002±0.001	0.000±0.000	0.000±0.000	0.000±0.000	0.002±0.000
CCS	3.683±0.642	4.723±0.513	4.704±0.530	3.293±0.409	3.680±0.423	3.223±0.379	2.988±0.358
EE - Cooling	1.044±0.223	1.255±0.267	1.273±0.256	0.749±0.122	1.045±0.221	0.749±0.134	0.667±0.110
EE - Heating	0.336±0.047	0.930±0.135	0.920±0.138	0.286±0.030	0.614±0.414	0.298±0.041	0.264±0.032
ETP	0.145±0.027	0.143±0.012	0.570±0.041	0.136±0.012	0.146±0.036	0.136±0.014	0.122±0.013
FF	21.292±14.003	20.451±13.099	19.287±13.101	14.717±12.093	19.683±12.202	15.540±12.505	20.117±11.955
PTHC	6.334±0.985	7.300±1.177	7.507±1.279	6.038±1.260	6.448±1.386	6.056±1.184	6.124±1.080
MI	98.237±16.045	86.002±14.692	86.133±14.969	55.108±12.026	64.788±14.652	63.155±13.998	52.273±11.612
PT	12.182±1.387	11.591±1.121	11.739±1.210	11.809±1.074	11.496±1.197	11.699±1.217	11.373±1.213
QAT	0.954±0.188	0.834±0.160	0.821±0.167	0.801±0.173	0.851±0.175	0.825±0.166	0.812±0.164
QFT	0.685±0.080	0.646±0.077	0.645±0.069	0.634±0.071	0.666±0.091	0.631±0.075	0.619±0.072
SP - M	1.366±0.333	1.164±0.210	1.155±0.241	1.070±0.186	1.113±0.191	1.094±0.174	1.089±0.187
SP - M/G3	3.213±0.638	3.032±0.562	3.095±0.571	3.000±0.543	2.984±0.576	2.979±0.591	3.009±0.608
SP - P	0.852±0.183	0.820±0.186	0.818±0.198	0.800±0.176	0.850±0.212	0.792±0.174	0.810±0.177
SP - P/G3	2.032±0.250	1.961±0.278	1.969±0.241	1.983±0.254	1.999±0.319	1.938±0.305	1.959±0.256
SM	0.003±0.001	0.004±0.001	0.122±0.012	0.000±0.000	0.003±0.001	0.001±0.001	0.001±0.000
TRP	15.275±1.287	14.410±1.055	14.362±1.019	11.963±0.932	13.722±0.972	12.795±1.252	10.388±0.926
YH	0.464±0.200	2.565±0.898	2.652±0.938	0.394±0.180	0.534±0.273	0.393±0.136	0.365±0.122

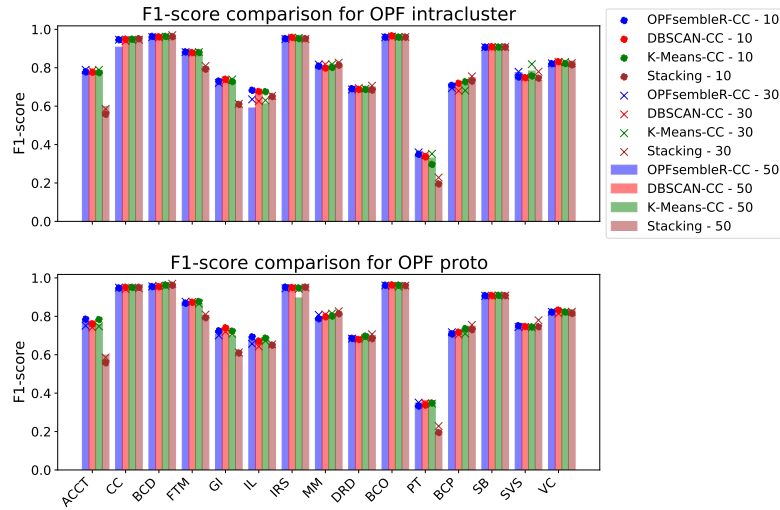


Fig. 5: Average F1-score obtained from each ensemble model in terms of the counting class diversity measure.

Table 2: Average MAE for each dataset and ensemble approach for 30 classifiers.

	30 classifiers						
	DBSCAN proto	DBSCAN intracluster	<i>k</i> -Means proto	<i>k</i> -Means intracluster	OPF proto	OPF intracluster	Stacking
ASN	2.847±0.298	2.379±0.355	3.113±0.195	1.181±0.152	2.513±0.448	1.186±0.137	1.000±0.109
DC - HX	0.001±0.000	0.001±0.000	0.002±0.001	0.000±0.000	0.000±0.000	0.000±0.000	0.001±0.000
DC - TX	0.001±0.000	0.000±0.000	0.003±0.001	0.000±0.000	0.000±0.000	0.000±0.000	0.001±0.000
CCS	7.186±1.118	5.489±0.574	5.346±0.632	3.250±0.433	4.378±0.687	3.375±0.455	2.901±0.394
EE - Cooling	1.718±0.598	1.550±0.304	1.528±0.298	0.743±0.115	1.704±0.357	0.766±0.154	0.688±0.109
EE - Heating	1.171±0.505	1.335±0.201	1.308±0.207	0.283±0.042	1.553±0.279	0.289±0.047	0.273±0.043
ETP	0.162±0.057	0.167±0.018	1.005±0.057	0.135±0.017	0.155±0.044	0.138±0.015	0.116±0.012
FF	21.819±19.362	18.027±13.164	17.692±12.115	14.419±12.266	18.933±12.438	14.707±12.362	26.785±11.686
PTHC	9.043±1.919	7.853±1.227	7.742±1.286	5.983±1.316	7.215±1.833	6.029±1.334	6.856±1.312
MI	108.094±17.841	90.743±14.768	88.415±14.921	51.680±11.603	96.375±15.336	51.960±10.788	52.179±13.003
PT	12.612±1.054	11.764±1.184	11.655±1.210	11.728±1.144	11.820±1.239	11.825±1.247	11.690±1.289
QAT	0.932±0.136	0.825±0.178	0.872±0.190	0.850±0.176	0.875±0.193	0.822±0.172	0.823±0.169
QFT	0.665±0.085	0.634±0.067	0.677±0.086	0.633±0.075	0.697±0.078	0.632±0.075	0.626±0.068
SP - M	1.582±0.341	1.276±0.253	1.319±0.257	1.108±0.190	1.210±0.266	1.064±0.172	1.102±0.228
SP - M/G3	3.506±0.596	3.065±0.624	3.083±0.633	2.926±0.565	3.036±0.611	2.976±0.579	3.057±0.555
SP - P	0.951±0.191	0.842±0.183	0.886±0.203	0.781±0.182	0.907±0.204	0.788±0.174	0.831±0.181
SP - P/G3	2.099±0.366	1.940±0.271	1.945±0.309	1.951±0.290	1.984±0.276	1.936±0.278	2.024±0.263
SM	0.017±0.004	0.012±0.002	0.286±0.024	0.000±0.000	0.016±0.002	0.000±0.000	0.001±0.000
TRP	9.698±1.372	14.339±1.032	14.214±1.009	7.184±0.877	14.139±1.095	7.074±0.960	7.102±0.922
YH	6.329±2.358	3.904±1.341	3.603±1.432	0.350±0.155	4.085±1.424	0.357±0.152	0.388±0.166

Overall, the results reveal that Stacking Generalization and *k*-Means stand out for lower average MAE with fewer classifiers, i.e., 10 and 30 models. The superiority stems from stacking robustness, which mitigates the impact of underperforming models when considering all base models. On the other hand, OPF Intracluster exhibits competitive performance across several scenarios, standing out as the best variant of the proposal for

Table 3: Average MAE for each dataset and ensemble approach for 50 classifiers.

	50 classifiers						
	DBSCAN proto	DBSCAN intracluster	k-Means proto	k-Means intracluster	OPF proto	OPF intracluster	Stacking
ASN	2.991±0.248	2.204±0.222	2.275±0.223	1.143±0.151	5.329±5.693	1.151±0.175	1.042±0.179
DC - HX	0.001±0.000	0.000±0.000	0.001±0.000	0.000±0.000	0.001±0.000	0.000±0.000	0.001±0.000
DC - TX	0.001±0.000	0.000±0.000	0.001±0.000	0.000±0.000	0.001±0.000	0.000±0.000	0.001±0.000
CCS	6.329±1.212	5.222±0.589	5.337±0.571	3.004±0.402	39.333±45.987	2.972±0.378	2.819±0.372
EE - Cooling	1.646±0.445	1.389±0.287	1.505±0.286	0.674±0.104	1.758±0.395	0.676±0.119	0.635±0.111
EE - Heating	1.509±0.400	1.100±0.172	1.455±0.231	0.294±0.031	1.482±0.466	0.274±0.031	0.269±0.036
ETP	0.208±0.068	0.166±0.019	1.667±0.436	0.136±0.014	0.584±0.060	0.131±0.015	0.112±0.014
FF	27.136±25.187	21.554±13.127	18.880±12.728	15.971±12.754	20.081±11.971	15.006±13.266	28.520±13.404
PTHC	9.628±1.862	7.864±1.371	7.868±1.182	6.121±1.275	7.139±1.246	6.021±1.239	6.538±1.439
MI	101.378±16.495	91.916±13.983	85.411±13.415	50.554±12.298	1840.808±3108.020	52.027±12.872	125.335±314.647
PT	11.949±1.196	11.630±1.119	11.594±1.202	11.774±1.190	11.653±1.130	11.475±1.112	11.658±1.288
QAT	0.850±0.173	0.833±0.168	0.916±0.201	0.829±0.171	0.836±0.176	0.833±0.177	0.834±0.194
QFT	0.660±0.094	0.625±0.068	0.792±0.133	0.629±0.069	0.641±0.076	0.621±0.069	0.624±0.084
SP - M	1.602±0.393	1.182±0.213	1.339±0.229	1.097±0.141	1.207±0.241	1.066±0.181	1.160±0.233
SP - M/G3	3.517±0.631	3.127±0.618	3.117±0.593	2.918±0.590	3.063±0.599	2.935±0.575	3.177±0.603
SP - P	0.928±0.178	0.800±0.175	0.996±0.227	0.800±0.172	0.832±0.181	0.797±0.174	0.838±0.188
SP - P/G3	2.018±0.293	1.927±0.271	1.944±0.291	1.921±0.264	1.948±0.271	2.002±0.248	2.048±0.291
SM	0.008±0.001	0.009±0.001	0.597±0.052	0.000±0.000	0.077±0.071	0.000±0.000	0.001±0.001
TRP	9.844±1.191	14.673±1.019	14.227±1.031	7.464±0.980	14.900±1.060	7.437±0.981	6.984±0.812
YH	6.448±2.311	3.375±1.063	3.425±1.274	0.363±0.232	2.925±0.955	0.313±0.172	0.352±0.121

the largest number of classifiers (50) and achieving the lowest MAE across 10 datasets. Such results indicate that focusing on the best-performing clusters reduces error propagation and maintains the technique’s predictive capacity when a larger number of classifiers is used in the ensemble. This behavior also suggests that the effectiveness of pruning depends on the size of the initial ensemble. When only a small number of classifiers is available, pruning may remove relevant models that contribute complementary predictions, reducing the benefit of model selection. However, as the number of classifiers increases, the ensemble tends to contain more redundant models with similar predictive behavior. In this scenario, pruning becomes more effective because OPFsembleR can remove redundant classifiers while preserving representative and diverse models.

Concerning the Forest Fires (FF) dataset, the method achieved an MAE of 15.006 ± 13.266 with 50 models (Table 3), a result superior to that of stacking (28.520 ± 13.404) and the lowest among all the baselines used for comparison. In the case of Parkinson’s Telemonitoring (PT), the results were also balanced among the three approaches tested, with OPF Intracluster recording 11.475 ± 1.112 for the highest number of classifiers in the ensemble, slightly better than stacking, which obtained 11.658 ± 1.288 in the same configuration. The same behavior is observed in three versions of the Student Performance (SP) datasets and Yacht Hydrodynamics (YH), in which OPFsemble attained lower MAE values. Additionally, the pruning strategy stands out in the QFT set for its consistency. It achieves an MAE of 0.621 ± 0.069 , which is comparable to all baselines across 50 models. However, for the TRP dataset, OPF Intracluster improved with more models than with 10 classifiers, but stacking still achieved the best MAE of 6.984 ± 0.812 . However, OPF intracluster still provides the lowest MAE among the counterpart clustering techniques. Regarding the DC databases, the results showed MAE values close to zero across all approaches, indicating the task’s simplicity for the models tested.

Finally, Figure 6 presents the critical differences from the Nemenyi test. Figure 6a) shows the ranks and critical differences for 10 classifiers, indicating that intracuster-based pruning ensembles are the best-performing models. In addition, OPF based on intracuster pruning achieved performance similar to k -Means-based intracuster pruning and stacking generalization, as indicated by the ranking and the horizontal bars connecting those models. Among 30 classifiers (Figure 6b), OPF with intracuster is the second-ranked model, with performance similar to k -Means and stacking, as highlighted by the connecting bar. Among 50 classifiers, intracuster-based OPF ranks first, followed by k -Means and stacking (Figure 6c), thus confirming its ability to handle pruning with a larger number of classifiers, as previously discussed. Intracuster-based strategies offer an alternative to ensemble pruning, in which the most effective models strike a balance between selecting models with diverse predictive capabilities, including those with weak performance, and selecting only those with higher forecasting abilities. This result reinforces that OPF-based pruning is particularly advantageous in larger ensembles, where redundancy among classifiers is more likely, and selecting representative models can reduce complexity without significantly harming predictive performance.

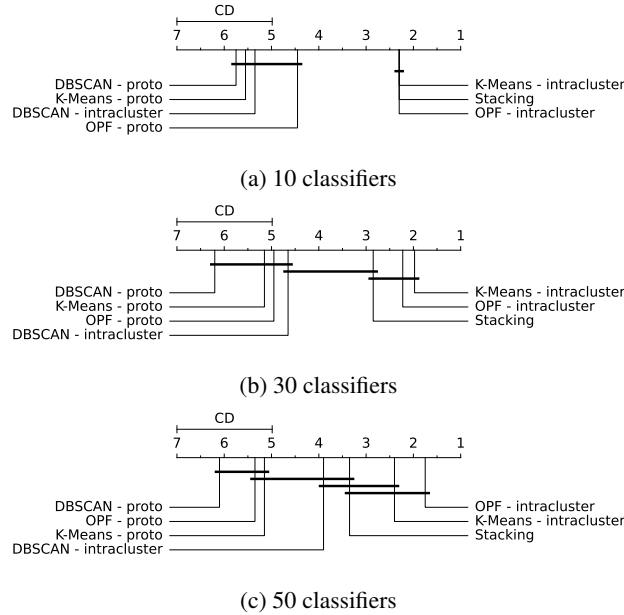


Fig. 6: Nemenyi test for each ensemble model.

Final Remarks The statistical test indicated similarities between k -means and OPF, but k -means requires a predetermined number of clusters, while OPF's k^* (or $k_{\max}$)

establishes an upper limit on local connectivity for density estimation without a fixed partition. In unsupervised OPF, $k_{\max}$ defines the maximum neighborhood size during estimation, and the actual number of clusters is determined by the local maxima of the probability density function [31]. This adaptive approach allows OPF to find clusters dynamically, making it suitable for data with small or irregular structures when prior information is lacking. A sufficiently large $k_{\max}$ reduces the sensitivity to further increases, as the primary connectivity structure of the data is already captured. Thus, $k_{\max}$ directly influences the size of the neighborhood rather than the number of clusters.

6 Conclusions

This paper presented the OPFsembleR, a classification and regression ensemble pruning framework based on the unsupervised Optimum-Path Forest algorithm. The proposed framework was extended to perform regression tasks. Additionally, three divergence measures were incorporated into the classification task to provide diversity across different metadata reduction methods. In most cases, the proposed method outperformed the stacked generalization ensemble, providing more accurate results in larger ensembles. Furthermore, the OPFsembleR leverages the advantages of OPF customization in several machine learning tasks, using fewer hyperparameters to model the data distribution. Future work could extend the pruning strategies to other ranking approaches for the selected models to assign importance based on prediction error. Although selecting a single prototype from each cluster was effective in preserving ensemble diversity and reducing redundancy, future studies will investigate the selection of additional models beyond the prototype. This direction may allow the ensemble to retain complementary classifiers within the same cluster, especially in cases where more than one model presents relevant predictive behavior.

Acknowledgments. This study was partially financed by the São Paulo Research Foundation (FAPESP), Brazil, grant numbers 2013/07375-0, 2019/07665-4, 2023/14427-8, 2023/10823-6, 2023/03726-4, 2023/12830-0, 2023/01374-3, and 2023/14354-0. The authors are also grateful to the Brazilian National Council for Scientific and Technological Development (CNPq) grant 308529/2021-9, to Unesp-IEPe-RC-#06/2023 PROPe grant, to the Petrobrás Brazil grant #2017/00285-6, to the Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) grant 88887.805805/2023-00, and to the Center for Research and Innovation on Smart and Quantum Materials (CRISQuaM) grant 24/00998-6.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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